Rekayasa Sistem Berbasis Pengetahuan

ETS

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- Tim Riset Bu Licha ingin mengembangkan prototipe Knowledge-Based Ergonomic Assessment System untuk membantu evaluator mengidentifikasi faktor risiko pada workstation pekerja kantor. Sistem menggunakan metode Rapid Office Strain Assessment (ROSA) untuk menilai kondisi kursi, monitor, telepon, keyboard, dan mouse.
- Evaluator mengamati workstation seorang staf administrasi yang bekerja menggunakan komputer setiap hari. Hasil observasi dicatat pada Tabel.

- Tugas Tim Anda setelah belajar RSBP Adalah sbb:
- Apa saja Fakta hasil assessment.
- Pilih knowledge representation yang menurut tim Anda paling sesuai, dan pertimbangannya.
- Aturan inferensi apa yang Anda buat untuk melakukan reasoning?
- Query untuk memperoleh skor final ROSA setiap staf.

Tabel pengukuran 10 staf kantor dengan ROSA Worksheet

Skor setiap komponen dan skor akhir ROSA harus ditentukan berdasarkan worksheet dan aturan skoring yang digunakan dalam perkuliahan. Durasi pada tabel merupakan jam penggunaan per hari.

Nama staf	Sudut lutut	Jarak duduka n–lutut	Armrest dan siku	Sudut sandaran	Monitor: jarak dan posisi	Telepon	Ekstensi pergelang an	Mouse	Durasi kursi / monitor / telepon / keyboard / mouse (jam/hari)
Melati	105°	1 inci	Siku terangkat; bahu terangkat	115°	80 cm; layar tinggi, leher ekstensi	35 cm; dijepit bahu	20°	Menjangkau; palmrest di depan mouse	8 / 7 / 2 / 7 / 7
Mawar	90°	3 inci	Siku tersangga; bahu rileks	100°	60 cm; setinggi mata	Headset; leher netral	5°	Dekat keyboard	7 / 7 / 1 / 7 / 7
Kanaka	85°	4 inci	Siku tidak tersangga; bahu rileks	90°	50 cm; layar rendah	35 cm; tanpa hands-free	18°	Menjangkau; permukaan berbeda	6 / 5 / 3 / 5 / 5
Seruni	95°	3 inci	Armrest terlalu rendah; lengan tidak tersangga	105°	85 cm; layar tinggi, leher ekstensi	Headset; leher netral	10°	Menjangkau mouse	8 / 8 / 1 / 8 / 8
Kenanga	110°	2 inci	Armrest terlalu tinggi; bahu terangkat	112°	65 cm; setinggi mata; tanpa document holder	25 cm; telepon di tangan	5°	Dekat keyboard; palmrest di depan mouse	5 / 4 / 2 / 4 / 4
Cempaka	90°	3 inci	Siku tersangga; bahu rileks	100°	55 cm; layar rendah	Headset; leher netral	20°	Menjangkau; permukaan berbeda	6 / 6 / 0,5 / 6 / 6
Teratai	92°	3 inci	Siku tersangga; bahu rileks	100°	45 cm; leher berputar 35°	40 cm; dijepit bahu	5°	Dekat keyboard	7 / 7 / 4 / 7 / 7
Padma	80°	5 inci	Armrest terlalu tinggi; bahu terangkat	90°	60 cm; setinggi mata; dokumen tanpa holder	Headset; leher netral	10°	Menjangkau; palmrest di depan mouse	8 / 8 / 2 / 8 / 8
Raflesia	100°	2 inci	Siku tidak tersangga; bahu rileks	115°	75 cm; layar rendah	35 cm; tanpa hands-free	22°	Menjangkau; pinch grip	5 / 5 / 2 / 5 / 5
Edelweis	90°	3 inci	Siku tersangga; bahu rileks	100°	60 cm; setinggi mata	Headset; leher netral	5°	Dekat keyboard	9 / 9 / 1 / 9 / 9

ROSA Worksheet

- ROSA adalah sistem penilaian yang membagi ergonomi kantor menjadi tiga bagian utama:
- A: Pengaturan Kursi – menutupi tinggi, ruang duduk, sandaran tangan, dan penyangga punggung
- B: Penggunaan Monitor dan Ponsel – posisi layar, penempatan ponsel, dan postur leher
- C: Penggunaan Keyboard dan Mouse – penyelarasan, posisi tangan, dan dukungan
-
- Setiap bagian menilai postur dan durasi. Semakin lama seseorang berada di posisi berisiko, semakin tinggi skornya—dan semakin tinggi risikonya.
- Hasil akhir ROSA adalah skor numerik dari 1 sampai 10:
- 1–3: Risiko rendah
- 4–5: Risiko sedang (beberapa perubahan diperlukan)
- 6 atau lebih tinggi: Risiko tinggi (tindakan segera diperlukan)

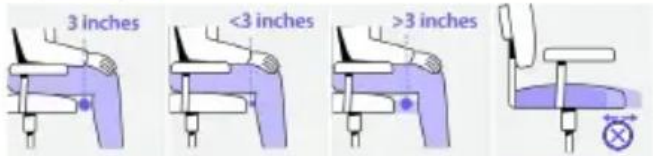
Section A – Chair

A.1 Chair height



Knees at 90° (+1) | Too low (knee angle <90°) (+2) | Too high (knee angle >90°) (+2) | No foot contact on ground (+3) | Insufficient Space Under Desk- No Ability to Cross Legs (+1) | Non-Adjustable (+1)

A.2 Pan Depth



Approximately 3 inches of space between knee and edge of seat (+1) | Too Long : Less Than 3" of space (+2) | Too Short : More than 3" of Space (+2) | Non-Adjustable (+1)

A.1 Score + A.2 Score = **Section A – Y axis Score**

A.3 Armrests



Elbows supported in line with shoulder, shoulders relaxed (+1) | Too High (Shoulders Shrugged)/Low (Arms Unsupported) (+2) | Hard/damaged surface (+1) | Too wide (+1) | Non-Adjustable (+1)

A.4 Back Support



Adequate Lumbar Support - Chair reclined between 95°-110° (+1) | • No Lumbar Support OR; • Lumbar Support not Positioned in Small of Back (+2) | • Angled Too Far Back (Greater than 110°) OR; • Angled Too far forward (Less than 95°) (+2) | No Back Support (ie Stool OR Worker Leaning forward) (+3) | Work Surface too High (Shoulders Shrugged) (+1) | Back Rest Non-Adjustable (+1)

A.3 Score + A.4 Score = **Section A – X axis Score**

Duration Instructions

- If less than 30 minutes continuously, or less than 1 hour per day : **-1**
- If between 30 minutes and 1 hour continuously, or between 1 and 4 hours per day : **0**
- If greater than 1 hour continuously, or more than 4 hours per day : **+1**

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SCORES

Section A Score	X Axis: Arm Rest and Back Support							
Y Axis: Chair height / Pan Depth	2	3	4	5	6	7	8	9
2	1	2	3	4	5	6	7	8
3	2	2	3	4	5	6	7	8
4	3	3	3	4	5	6	7	8
5	4	4	4	4	5	6	7	8
6	5	5	5	5	6	7	8	9
7	6	6	6	7	7	8	8	9
8	7	7	7	8	8	9	9	9

Duration + Section A = **Chair ROSA Score**

Section B Score	B.1 Score							
B.2 Score	0	1	2	3	4	5	6	7
0	1	1	1	2	3	4	5	6
1	1	1	2	2	3	4	5	6
2	1	2	2	3	3	4	6	7
3	2	2	3	3	4	5	6	8
4	3	3	4	4	5	6	7	8
5	4	4	5	5	6	7	8	9
6	5	5	6	7	8	8	9	9

Secion B Score

Section C Score	C.1 Score							
C.2 Score	0	1	2	3	4	5	6	7
0	1	1	1	2	3	4	5	6
1	1	1	2	3	4	5	6	7
2	1	2	2	3	4	5	6	7
3	2	3	3	3	5	6	7	8
4	3	4	4	5	5	6	7	8
5	4	5	5	6	6	7	8	9
6	5	6	6	7	7	8	8	9
7	6	7	7	8	8	9	9	9

Secion C Score

Monitor & Peripherals ROSA	Section C Score								
Section B Score	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	2	3	4	5	6	7	8	9
3	3	3	3	4	5	6	7	8	9
4	4	4	4	4	5	6	7	8	9
5	5	5	5	5	5	6	7	8	9
6	6	6	6	6	6	6	7	8	9
7	7	7	7	7	7	7	7	8	9
8	8	8	8	8	8	8	8	8	9
9	9	9	9	9	9	9	9	9	9

Monitor & Peripherals ROSA SCORE

Section B – Monitor and Telephone

B.1 Monitor

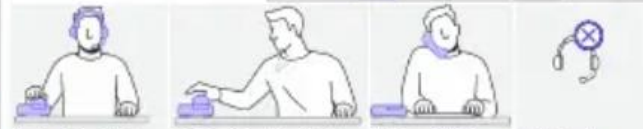
Duration + Monitor = **B.1 Score**



Arm's Length Distance (40-75cm) / Screen at Eye Level (+1) | • Too Low (below 30°) (+2) • Too Far (+1) | Too High (Neck Extension) (+3) | Neck Twist Greater than 30° (+1) | Glare on Screen (+1) | Documents - No Holder (+1)

B.2 Phone

Duration + Phone = **B.2 Score**



Headset / One Hand on Phone & Neutral Neck Posture (+1) | Too Far of Reach (outside of 30cm) (+2) | Neck and Shoulder Hold (+2) | No Hands-Free Options (+1)

Section C – Mouse and Keyboard

C.1 Mouse

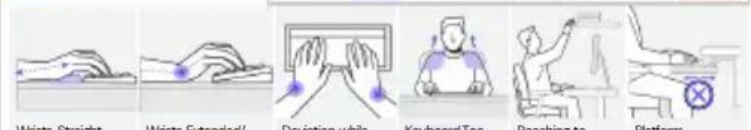
Duration + Mouse = **C.1 Score**



Mouse in Line with Shoulder (+1) | Reaching to Mouse (+2) | Mouse/Keyboard on Different Surfaces (+2) | Pinch Grip on Mouse (+1) | Palmrest in Front of Mouse (+1)

C.2 Keyboard

Duration + Keyboard = **C.2 Score**



Wrists Straight, Shoulders Relaxed (+1) | Wrists Extended/ Keyboard on Positive Angle (>15° Wrist extension) (+2) | Deviation while Typing (+1) | Keyboard Too High : Shoulders Shrugged (+1) | Reaching to Overhead Items (+1) | Platform Non-Adjustable (+1)

ROSA FINAL	Monitor & Peripherals ROSA									
Chair ROSA	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	2	3	4	5	6	7	8	9	10
3	3	3	3	4	5	6	7	8	9	10
4	4	4	4	4	5	6	7	8	9	10
5	5	5	5	5	5	6	7	8	9	10
6	6	6	6	6	6	6	7	8	9	10
7	7	7	7	7	7	7	7	8	9	10
8	8	8	8	8	8	8	8	8	9	10
9	9	9	9	9	9	9	9	9	9	10
10	10	10	10	10	10	10	10	10	10	10

ROSA Final Score

TuMeke Ergonomics

based on THE RAPID OFFICE STRAIN ASSESSMENT DEVELOPED BY MICHAEL SONNE, MHK, CK.